

# CH1 數與式

1.  $\frac{353}{990}$  2. (B)(C)(D) 3.8 或  $-16$  4.  $\sqrt{3}$  5.  $-4 \leq x \leq 5$  6.  $x$  為任意實數 7.  $B <$

$A < C$  8.450 9.57 10.63

解析

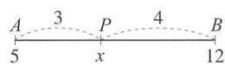
$$1. 0.\overline{12} + 0.2\overline{35} = \frac{12}{99} + \frac{235-2}{990} = \frac{12}{99} + \frac{233}{990} = \frac{120+233}{990} = \frac{353}{990}$$

2. (A) 令  $a = \sqrt{2}, b = \sqrt{2}, \frac{\sqrt{2}}{\sqrt{2}} = 1$  為有理數

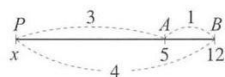
(B)  $a = \frac{4(2a+b)-2(2b+c)+(2c+a)}{9}$  為有理數同理  $b$ 、 $c$  亦為有理數

(E) 令  $a = \sqrt{3}, b = -1$ , 則  $a + b\sqrt{3} = \sqrt{3} + (-1)\sqrt{3} = 0$  但  $a$ 、 $b$  均不為 0

故選(B)(C)(D)



圖(u)



圖(二)

3. (1)  $P$  為內分點時，如右圖(-)

$$x = \frac{3 \cdot 12 + 4 \cdot 5}{3 + 4} = \frac{56}{7} = \underline{8}$$

(2)  $P$  為外分點時，如右圖(二)

$$5 = \frac{1 \cdot x + 3 \cdot 12}{3 + 1}, 20 = x + 36 \text{ 則 } x = -16$$

$$4. \sqrt{4 + \sqrt{12}} = \sqrt{3 + 2\sqrt{3} + 1} = \sqrt{(\sqrt{3} + 1)^2} = \sqrt{3} + 1 = 2 + (\sqrt{3} - 1)$$

$$\therefore a = 2, b = \sqrt{3} - 1$$

$$\frac{1}{b} + \frac{1}{a+b} = \frac{1}{\sqrt{3}-1} + \frac{1}{2+(\sqrt{3}-1)} = \frac{\sqrt{3}+1+\sqrt{3}-1}{(\sqrt{3}-1)(\sqrt{3}+1)} = \sqrt{3}$$

5.  $|1-2x| \leq 9, |2x-1| \leq 9$  ,同除以 2 得

$$\left|x - \frac{1}{2}\right| \leq \frac{9}{2}, -\frac{9}{2} \leq x - \frac{1}{2} \leq \frac{9}{2}$$

$$-\frac{9}{2} + \frac{1}{2} \leq x \leq \frac{9}{2} + \frac{1}{2}, -4 \leq x \leq 5$$

6. (1) 當  $x - \frac{3}{4} > 0 \Rightarrow 2\left(x - \frac{3}{4}\right) > x - \frac{9}{2}$  ,即  $x > -3$  ,得  $x > \frac{3}{4}$

(2) 當  $x - \frac{3}{4} \leq 0 \Leftrightarrow 2\left(-x + \frac{3}{4}\right) > x - \frac{9}{2}$  ,即  $x < 2$  得  $x \leq \frac{3}{4}$

由(1)、(2)得  $x$  , 為任意實數

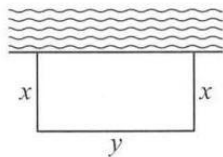
$$7. A^2 = 39 = 20 + 19$$

$$B^2 = (\sqrt{15} + \sqrt{5})^2 = 15 + 2\sqrt{15} \cdot \sqrt{5} + 5 = 20 + \sqrt{300}$$

$$C^2 = (\sqrt{13} + \sqrt{7})^2 = 13 + 2\sqrt{13} \cdot \sqrt{7} + 7 = 20 + \sqrt{364}$$

$$\therefore \sqrt{300} < 19 = \sqrt{361} < \sqrt{364}$$

$\therefore B^2 < A^2 < C^2$  ,即  $B < A < C$



8. 如右圖,設三邊長為  $x$ 、 $x$ 、 $y$  公尺

則  $2x + y = 60$ , 由算幾不等式知

$$\frac{2x + y}{2} \geq \sqrt{2xy} \Rightarrow 30 \geq \sqrt{2xy}$$

$$\Rightarrow xy \leq \frac{900}{2} = 450$$

當  $2x = y = 30$  時,面積  $xy$  有最大值 450 平方公尺

9. 令經過  $a$  分鐘後細菌總數會達到  $N$  個

已知  $1 \times 3^{60} = N$  及  $27 \times 3^a = N$

則  $3^{60} = N = 3^3 \times 3^a = 3^{3+a} \Rightarrow a = 60 - 3 = \underline{57}$  (分鐘)

10.  $\log E(8.8) = 1.5 \times 8.8 + 11.8 \Rightarrow E(8.8) = 10^{1.5 \times 8.8 + 11.8}$   $\log E(7.6) =$

$$1.5 \times 7.6 + 11.8 \Rightarrow E(7.6) = 10^{1.5 \times 7.6 + 11.8} \Leftrightarrow \frac{E(8.8)}{E(7.6)} = \frac{10^{1.5 \times 8.8 + 11.8}}{10^{1.5 \times 7.6 + 11.8}} = 10^{1.5 \times 1.2} =$$

$$10^{1.8} = 10 \times 10^{0.8} \approx 10 \times 6.31 \approx 63.1, \text{ 約為 } 63 \text{ 倍}$$